MIL-B-19906A(WEP)
1 AUGUST 1961
SUPERSEDING
MIL-B-19906(AER)
1 MAY 1957

MILITARY SPECIFICATION

BOMB RACK AND ROCKET LAUNCHER, COMBINATION, AERO 15 SERIES

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

* 1.1 Scope - This specification covers the manufacture, inspection and testing of one type of combination bomb rack and rocket launcher designated as Bomb Rack and Rocket Launcher, Combination, Aero 15 Series, and hereinafter referred to simply as rack.

2. APPLICABLE DOCUMENTS

* 2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-7743	Testing, Store Suspension Equipment, General Specification for
MIL-P-7936	Parts and Equipment Aeronautical, Preparation for Delivery
MIL-B-21034	Bomb Arming Unit Assemblies, Aero 2A-1, 2A-2, 2A-3 and 2B
MIL-H-22161	Hook, Bomb Rack, Aero 15 Series
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1095

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MIL-STD-143	Specification and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

DRAWINGS

Bureau of Naval Weapons

55A82ASK-1	Drawing Index, Bomb Rack and Rocket Launcher Combination, Aero 15B
55A154ASK-1	Drawing Index, Bomb Rack and Rocket Launcher Combination, Aero 15C
55A154ASK-7	Drawing Index, Bomb Rack and Rocket Launcher Combination, Aero 15C-1
60A119-1	Drawing List, Bomb Rack and Rocket Launcher, Combination Aero 15D
58A154ASK-4	Drawing Index, Bomb Arming Unit Aero 2A-1
563178	5.0-Inch Rocket Motor Mk 10 Mod 6 Button Lug Band
563185	5.0-Inch Rocket Motor; Mk 10 Mod 6 Tunnel Lug Band
655885	5.0-Inch Rocket Mk 28 Mod 4 (Aircraft General Purpose); Outline Drawing
1283673	Mk 25 Mod 1 Primer

* 2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(2) Applicable drawings may be obtained upon request to the Commanding Officer, Naval Supply Depot, Code CDD, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - Parts and materials used in the manufacture of the rack shall conform to the applicable specifications as specified herein or as listed on the applicable drawings. Material conforming to contractors' specifications may be used, provided the specifications are approved by the procuring agency, are equivalent to the Government specifications and contain requirements for adequate tests of the material. The use of contractors' specifications shall not constitute a waiver of Government inspection. All materials shall be of the lightest practical weight and entirely suitable for the intended purpose. When materials are used which are procured under commercially recognized, non-Government specifications, a certificate acceptable to the Government Inspector, showing compliance with such specifications, shall be furnished by the supplier of such material.

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* 3.1.1 Critical and Strategic Materials - Materials shall be chosen on the basis of suitability and availability in this country in quantity and from consideration of the additional restrictions created during a national emergency. Critical and strategic materials shall be used only to the extent considered mandatory for satisfactory operation. The contemplated use of these materials shall be brought to the attention of the procuring agency with appropriate justifications therefor and shall be subject to approval of the procuring agency.

* 3.1.2 Selection of Materials - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with MIL-STD-143.

* 3.1.2.1 Standard Parts - Standard parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN or JAN) without alteration and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawing. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used, provided they conform to all requirements of this specification.

3.1.2.2 Weight Reduction - The contractor shall make every effort to reduce the equipment's over-all weight without impairing the function, operation or durability. All changes that are required to accomplish weight reduction shall be subject to written approval by the procuring agency prior to incorporation.

* 3.1.3 Hook, Bomb Rack, Aero-15 Series - Hooks conforming with Specification MIL-H-22161 shall be used with the rack. The combination of rack and hooks shall meet all requirements of this specification satisfactorily.

* 3.1.4 Arming Assemblies - Bomb arming unit assemblies conforming to Specification MIL-B-21034 shall be used with the Bomb Rack Aero -15 Series. The combination of arming assembly and bomb rack shall meet all applicable requirements of this specification.

3.2 Design and Construction -

* 3.2.1 Conformance to Drawings and Specifications - The Bomb Rack and Rocket Launcher, Combination, Aero-15 Series shall be constructed in accordance with the applicable drawings, specified in 2.1. The purpose of the performance requirements specified in Section 4 is to verify that the contractor has complied with such drawings. In the event that the contractor ascertains during the performance of the contract that compliance with such drawings will result in production of equipment that will not meet all the requirements of Section 4, the contractor shall promptly notify the procuring agency of such discrepancy. No departure from or revision to the drawings or this specification shall be made without written approval from the procuring agency.

* 3.2.2 Input Power Requirements - This equipment shall perform satisfactorily under all applicable conditions specified in Specification MIL-T-7743, from a 28 volt power source with characteristics and limits as defined in MIL-STD-704.

* 3.2.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.3 Workmanship - All details of workmanship shall be in accordance with high grade armament manufacturing practice for the quantity production of this type of equipment. All parts shall be free from excessive marks, grooving, indentations, cracks, flaws and any other injurious defects which may impair the serviceability of the assembled equipment. All electrical components and wiring shall be installed in accordance with the workmanship requirements of Specification MIL-E-5400. All workmanship shall be of a quality satisfactory to the procuring agency.

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3.3.1 Manufacturing and Finish - Manufacturing processes and finishes shall be as shown on the applicable drawings specified in 2.1 unless otherwise approved by the procuring agency, (see 3.2.1). The responsibility shall be assumed by the contractor for any malfunction or failure of the parts of the assembled equipment to pass tests specified herein due to unsatisfactory fits, clearance, tolerances, internal stresses or finishes.

* 3.4 Correction of Defects - Parts not supplied by the Government shall be considered contractor's responsibility. Any defects in items, parts and materials discovered during the inspection and testing shall be corrected by the contractor at his expense. Such corrective measures as may be found necessary shall likewise be incorporated in all subsequent bomb racks offered for acceptance under the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements:

(1) Preproduction Tests - (4.2)

(2) Acceptance Tests - (4.3)

* 4.1.1 Test Samples - The equipments used for preproduction and sampling tests shall not be considered as equipments under the contract. They are to be supplied as equipments over and above the quantity specified for delivery in the contract. Samples may be returned to the contractor or retained at the option of the procuring agency.

* 4.1.2 Test Conditions - The test conditions for all tests shall be as defined in Specification MIL-T-7743, unless otherwise specified.

* 4.2 Preproduction Tests - Preproduction tests shall be accomplished at a Government selected laboratory under the responsibility of the procuring agency. Five samples shall be selected for preproduction tests. These samples shall be the first five equipments produced using production tools. Prior to selecting these five samples for preproduction test each shall successfully pass the tests specified for individual tests. These tests shall be conducted by the contractor and results of these tests shall accompany the five samples sent to the Government selected laboratory for preproduction tests. Preproduction tests shall consist of all the tests specified in 4.4 and conducted in the sequence of Table I. Any defects or failures occurring during these tests shall be cause for tentative rejection of the preproduction samples. All rejected samples will be returned to the contractor for rework or correction of defects when considered practical and feasible by the procuring agency, otherwise the contractor will be required to submit additional samples for retest in lieu of the reworked samples.

4.2.1 Preproduction Approval - Approval of the preproduction samples shall be by the procuring agency upon satisfactory completion of all tests. No production equipments or lot samples shall be delivered prior to approval of the preproduction samples. Prefabrication of production equipment prior to the approval of the preproduction models is at the contractor's own risk. The approved preproduction samples will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance.

4.3 Acceptance Tests - Acceptance of equipment shall be on a lot basis. Each lot shall be composed of equipment which has successfully passed the individual tests. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

* 4.3.1 Individual Tests - Individual Tests shall be accomplished, under the responsibility of the contractor, on each equipment submitted for acceptance under the contract. The contractor shall maintain test reports showing the quantitative results for all individual tests. Such reports shall be signed by an authorized representative of the contractor and this data shall be made available to the procuring agency upon request. Individual tests shall consist of the following tests:

(1) Inspection of Product (4.4.1)

(2) Operational Tests (4.4.2)

(3) Rocket Installation Test (4.4.3)

(4) Rocket Release Mechanism Test (4.4.4)

(5) Launcher Latch Test (4.4.5)

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(6) Arming Mechanism Test (4.4.7)

(7) Electrical Fuse Arming Mechanism Test (4.4.8)

*4.3.2 Sampling Tests - Sampling tests shall be conducted at a government laboratory under the responsibility of the procuring agency. Sampling tests shall be conducted on samples selected from lots of equipment which have successfully passed the individual tests. Samples to be tested shall be selected at random by the government inspector in accordance with the following lot schedule and forwarded to the government selected laboratory for testing. A lot shall not exceed 300 racks except with the approval of the procuring agency. Samples shall be suitably labeled to identify them with a particular lot:

Lot Sizes of Equipment Offered for Acceptance	Quantity of Samples per Lot to be Tested
1 to 99	2
100 to 199	3
200 to 300	4

Sampling tests shall consist of the following tests performed in the sequence of Table I, unless specified otherwise by the procuring agency:

(1) Inspection of Product (4.4.1)

(2) Operational Test (4.4.2)

(3) Rocket Installation Test (4.4.3)

(4) Rocket Release Mechanism (4.4.4)

(5) Launcher Latch Test (4.4.5)

(6) Emergency Jettisoning Mechanism (4.4.6)

(7) Arming Mechanism (4.4.7)

(8) Electrical Fuse Arming Mechanism (4.4.8)

(9) Vibration Test Bomb Portion, Rocket Portion (4.4.9)

(10) Impact Test (4.4.10)

(11) Life Test Bomb Portion, Rocket Portion (4.4.11)

(12) Static Tests (Loads given in Para. 4.5, this spec.) Bomb Portion, Rocket Portion (4.4.12)

(13) Physical Characteristics (Bomb and Rocket Portion) (4.4.13)

(14) Manual Operation (4.4.14)

(15) Electrical Characteristics (Bomb and Rocket Portions) (4.4.15)

(16) Lag Tests (4.4.16)

(17) Shock Test (4.4.17)

(18) Hi-G Release Test (4.4.18)

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*4.3.3 Approval of Production Lots - Approval of production lots shall be by the procuring agency upon satisfactory completion of all Sampling tests. Should a failure occur during this testing, it shall be cause for rejection of that particular lot. If any defects appear or failures occur, the manufacturer shall be required to eliminate the faults by reworking the production lot. The reworked production lot shall again pass the tests of paragraph 4.3.1, after which the required number of samples shall be submitted for retests to be conducted in accordance with paragraph 4.3.2. If further failures occur during the First Retest, additional retests shall be at the discretion of the procuring agency. Inspection and approval of materials in the process of manufacture shall not be construed as insuring the acceptance or approval of the finished product.

4.4 Test Methods

4.4.1 Inspection of Product - Each rack shall be inspected to insure that it has been properly assembled and adjusted, that the electrical connections and circuits are correct, positive and secure, and that the workmanship and finish are in accordance with this specification.

4.4.2 Operational Test - Each completely assembled rack shall be mounted in a horizontal attitude on a test stand which closely simulates the mounting of the rack on an aircraft. The rack shall be latched (unloaded) and a safetying pin inserted through the safetying pin hole in the rack frame. The rack shall be released manually and then electrically to insure that the hooks remain latched. Upon removal of the safetying pin, the hooks shall fully open. The hooks shall be relatched and without inserting the safetying pin, the manual release shall be actuated to fully open the hooks. This operation shall be repeated three times. The rack shall then be loaded with an inert 500 pound bomb and the release circuit energized electrically with a 28 volt DC, 0.020 second pulse to release the bomb. This operation shall be repeated three times. After the third release, the bomb shall be loaded once again and the sway braces tightened down firmly against the bomb in accordance with the sway brace instruction decalcomania. A check shall be made to insure that the sway braces firmly restrict motion of the bomb.

4.4.3 Rocket Installation Test - An installation test of the button lug band (Dwg. 563178) and the tunnel lug band (Dwg. 563185) shall be made by mounting the bands on a dummy rocket and installing in the rack to insure proper fit under conditions of cumulative manufacturing tolerances permitted by the drawings.

4.4.4 Rocket Release Mechanism Test - A dummy rocket simulating the required rocket shall be installed on the rack and the rocket release solenoid energized. The dummy rocket shall then be removed from the rack without failure of the shear pin or other parts of the release mechanism.

*4.4.5 Launcher Latch Test - With a dummy rocket simulating the largest size rocket specified herein, the following tests shall be made:

(1) The shear pin shall shear with a 1500-pound (plus or minus 150 pounds) force, applied horizontally forward through the center of gravity of the dummy rocket with the rocket latch solenoid in the locked position.

(2) The rocket latch shall be tested to assure that it will hold a 200-pound force applied horizontally forward through the center of gravity of the dummy rocket with the rocket latch solenoid in the released or unlocked position.

*4.4.6 Emergency Jettisoning Mechanism Test - An inert 500-pound bomb, Mk 82 Mod 1, and a dummy rocket simulating the required rocket, drawing 655885, shall each be released from each rack by means of the emergency jettisoning mechanism, utilizing a Mark 25, Mod 1 primer. The circuit continuity for firing the Mark 25 Mod 1 primer, shown in drawing 1283673, shall be checked by a suitable indicating means. At least one jettison of a bomb and one jettison of a rocket shall be made on each sample, and on at least one sample, five bomb jettisons and five rocket jettisons shall be made. An emergency jettison clip, drawing 55A82A132, shall be installed for each bomb jettison test. After each firing the jettison barrel shall be cleaned with a suitable solvent and the lead coil and "O" ring replaced.

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* 4.4.7 Arming Mechanism Test - A bomb arming wire, Mk 1 (AN-M6A2), shall be inserted in each of the arming mechanisms and the arming solenoids individually energized. Apply a proof load of 100 pounds horizontally-forward and vertically-downward on the nose arming mechanism. Each load shall be applied separately through the arming wire. The arming wire swivel loop shall not pull out of the retainer nor shall any damage to the item occur. With the arming solenoid de-energized, the force required to withdraw the arming wire shall not be less than 2 pounds nor more than 8 pounds for any direction of pull in the forward quadrant for the nose arming mechanism and in the aft quadrant for the tail arming mechanism. These tests shall be conducted at room temperature.

* 4.4.8 Electrical Fuse Arming Mechanism Test - (when applicable)

(1) Cable Tension Test - A pull of 1 pound, minimum, exerted on the arming plug shall not extend the plug and cable from their fully retracted position. A pull of 15 ± 2.5 pounds shall extend the plug and cable fully. Upon the removal of this force, the plug and cable shall return to the fully retracted position.

(2) Operational Test - A minimum of five bomb releases shall be made with the applicable arming unit, (described in MIL-B-21034) connected to the bomb. The arming unit circuit shall be monitored in order to determine whether the prescribed electrical arming impulse is transmitted to the bomb.

(3) Electrical Continuity Checks - The necessary electrical continuity checks will be made to see that the arming unit circuit, in each state of its operational cycle, conforms to the electrical circuit and operation sequence of Figure 1.

* 4.4.9 Vibration Test - The vibration tests specified in Specification MIL-T-7743 shall apply for both the rocket portion and the bomb portion of the rack, except during the loaded portion of the test as follows:

(1) The rocket store shall not be sway braced.

(2) When only one rack is tested in vibration, the rack shall be vibrated for a period of 45 minutes, in each axis in turn, while loaded with either a bomb of maximum weight specified or a rocket of maximum size specified.

* 4.4.10 Impact Test - The launcher shall be secured to the carriage of an impact machine in a manner which permits the installation of a dummy rocket, nose down and perpendicular to the impact plane. The launcher shall be in a latched condition, and the assembly shall be subjected to increasing transient deceleration forces until release is effected. The release of the dummy rocket, prior to the failure of the latch shear pin, shall be considered failure of the unit to pass the test.

* 4.4.11 Life Test - (Rocket Portion) The launcher shall be latched and operated electrically for 500 cycles of operation at the minimum voltage. The rate of conducting the test shall be such as not to overheat the magnetic coils. The time interval between releases shall not be less than 30 seconds. The launcher shall be required to launch up to 200 live rockets without malfunction.

(1) Bomb Portion - The life test for the bomb portion shall be conducted as specified in Specification MIL-T-7743 except that the time interval between releases shall not be less than 30 seconds.

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* 4.4.12 Static Tests - The static tests for both rocket and bomb portions of the rack shall conform to Specification MIL-T-7743 with the following modifications:

(1) Rocket Portion - The dummy stores shall not be sway braced and the loads to be applied shall be as specified in 4.5.3.

(2) Bomb Portion - The loads of 4.5.2 shall be applied for the static tests on the bomb portion of the rack.

* 4.4.13 Physical Characteristics - The physical characteristics of the rack shall be checked as specified in Specification MIL-T-7743.

* 4.4.14 Manual Operation - The rack shall be tested for manual operation as specified in Specification MIL-T-7743 .

* 4.4.15 Electrical Characteristics - The electrical components for both the bomb and rocket portion of the rack shall be tested in accordance with Specification MIL-T-7743, with the following modification:

(1) Dielectric breakdown between layers - The unit shall have an emf of at least 166 percent of rated voltage applied across the coil. This circuit shall be made and broken rapidly 10 times with a minimum "on" time of 0.5 second. After the above operation and after cooling, the unit shall not show a change in resistance in excess of 5% while the ambient temperature is controlled to within ±5°F. The instruments used in this test shall be accurate within 1 percent.

* 4.4.16 Lag or Time of Release Test - The lag or time of release determination test shall be as specified in Specification MIL-T-7743.

* 4.4.17 Shock Test - The shock test of Specification MIL-T-7743 shall apply for the bomb portion of the rack except as modified by 4.5.

* 4.4.18 Hi-G Release Test - The rack shall meet the Hi-G tests of Specification MIL-T-7743.

4.5 Rack Test Loads

* 4.5.1 Shock Test Loads, Bomb Portion - The dynamic tests specified in Specification MIL-T-7743 shall be modified where necessary so that the tests are accomplished in each of the six directions parallel to the three principal axes of the rack. The load factors (G values) when the rack is carrying a 500 pound G.P. bomb (Mk 82 Mod 1) shall be according to the following table. The center of gravity of the bomb is approximately 6.38 inches below the bearing surfaces of the lugs and 7.00 inches aft of the forward lug.

Direction	Load Factors (G's)
Vertical Down	12.7
Vertical Up	7.2
Longitudinal Forward	9.9
Longitudinal Aft	9.9
Port	8.3
Starboard	8.3

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4.5.2 Static Test Loads (Bomb Portion) - The following are the critical limit inertial loads on the rack when carrying the 500 pound G. P. bomb (AN-M64A1). The center of gravity of the bomb is 7.93 inches below the bearing surfaces of the lugs and 7.0 inches aft of the forward lug.

Condition	V - pounds	S - pounds	D - pounds	PM - inch pounds	Y. M. - inch lbs.
Flight (c)	-3360	-4390	1020	9580	-12850
Flight (d)	-1980	-3120	-840	-23200	12850
Arrested Landing	-1500	0750	5000	4850	-2420

NOTES: (1) Sign Convention for Static Test Loads

V - up loads are positive

S - loads to the right when looking aft are positive

D - forward loads are positive

P. M. - nose up pitching moments are positive

Y. M. - nose right when loading aft yawing moments are positive

(2) Design yield loads are 1.15 times and ultimate loads are 1.50 times the limit values in the tables.

(3) All loads and moments are at the center of gravity of the store.

* 4.5.3 Static Test Loads (Rocket Launcher Portion) - The loads in the following table are the critical limit inertial loads of the 5-inch HVAR rocket applied at the center of gravity of the rocket when suspended on the rack. The tunnel lug band and the button lug band are positioned on the rocket so that the center of gravity of the rocket falls midway between their center lines.

Condition	V - pounds	S - pounds	D - pounds	P. M. - inch pounds	Y. M. - inch lbs.
Flight (a)	-1610	±210	±280	±1273	0
Flight (b)	-840	±1050	±280	±1273	0
Arrested Landing	-420	±210	±1400	±2545	±1273
Catapult	-350	±210	-1260	±2545	±849

The Notes of 4.5.2 apply to this paragraph also.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Unit Packaging -

* 5.1.1 Level A - Each rack shall be preserved and packaged in accordance with the requirements of Specification MIL-P-7936 Level A utilizing method 1A of Specification MIL-P-116, without the use of preservative compound. Each package shall contain one complete rack.

* 5.1.2 Level C - Preservation and Packaging shall be in conformance with Specification MIL-P-7936, Level C.

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* 5.2 Packing - The unit package shall be preserved as indicated in 5.1 and shall be packed in accordance with Specification MIL-P-7936, Level A, B or C, as specified in the contract or order. (See 6.5)

5.3 Marking and Labeling

5.3.1 Unit Packages - Each unit package shall be durably and legibly marked or labeled with the following information and in accordance with the requirements of MIL-STD-129.

Bomb Rack and Rocket Launcher, Combination, Aero 15 Series

Military Specification No. MIL-B-19906A(WEP)

Preserved (Date)

Domestic or Export Packed (as applicable)

Government Order No. (or Contract No. if order No. is not assigned).

Names of Contractors (and name of Manufacturer if not the same).

5.3.2 Shipping Containers - All shipping containers shall be marked in accordance with applicable requirements of MIL-STD-129.

6. NOTES

6.1 Intended Use - The rack is to be used in level bombing, dive bombing and other types of bombing and shall provide for mounting either externally as on a wing or fuselage or, internally as in a bomb bay. The rack shall be capable of carrying and releasing stores weighing up to 500 pounds and with bomb lug spacing of 14 inches. Also, it shall be capable of carrying and launching HVAR rockets and other rockets from rocket packages. The release circuits shall be capable of being operated electrically.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced herein. All referenced specifications shall apply to the extent specified.

6.3 Availability of Dummy Stores for Testing - The contractor shall submit requests to the procuring agency for dummy and inert stores to be used for testing as required by this specification.

6.4 Information Required with Bids - Each Bid shall be accompanied by a statement as to whether or not any exceptions are being taken to the requirements shown in the drawings and specifications. If exceptions are being taken or deviations are anticipated, alternative proposals shall be indicated in the statement and submitted therewith.

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* 6.5 Ordering Data - Procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Type of rack and components required.

(3) Level of preservation and packaging

(4) Level of packing

(5) Whether rough handling tests be performed on unit or exterior container

* 6.6 An asterisk preceding a paragraph number denotes a change in that paragraph from the basic specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely-related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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TABLE I

TEST SCHEDULE

BOMB RACK AND ROCKET LAUNCHER, COMBINATION, AERO 15

Preproduction and Production Tests which are to be performed on the Racks by the Testing Agency

Combination Under Test	Appl. Para.	Description of Tests	Lot Size	Sample Identification and Order of the Tests 1-99		100-199			200 to 300			
			Sample	A	B	A	B	C	A	B	C	D
Bomb and Rocket Portions	4.4.1	Inspection of Product		1	1	1	1	1	1	1	1	1
	4.4.2	Operational Test		2	2	2	2	2	2	2	2	2
	4.4.3	Rocket Installation Test		3	3	3	3	3	3	3	3	3
	4.4.4	Rocket Release Mechanism Test		4	4	4	4	4	4	4	4	4
	4.4.5	Launcher Latch Test		5	5	5	5	5	5	5	5	5
	4.4.6	Emergency Jettisoning Mechanism Test		6	17	6	7	12	6	11	6	9
	4.4.7	Arming Mechanism Test		7	6	7	8	7	7	6	7	7
	4.4.8	Electrical Fuse Arming Mechanism Test		8	7	9	9	8	8	7	8	8
Rocket Portion	4.4.9	Vibration Test		10a		10a			10			
	4.4.10	Impact Test			15		14					12
	4.4.11	Life Test			13			9			13	10
	4.4.12(1)	Static Test			14			11			14	
Bomb Portion	4.4.13	Physical Characteristics		9	8	8	6	6	9	8	9	6
	4.4.14	Manual Operation			11		10				10	
	4.4.15	Electrical Characteristics		12				13	11			
	4.4.16	Lag Tests			12		11				11	
	4.4.9	Vibration Test		10b		10b				10		
	4.4.17	Shock Test			16		15					13
	4.4.18	Hi-G Release Test			9	11	12			9		
	4.4.11(1)	Life Test			10		13				12	11
	4.4.12(2)	Static Test			11			10			15	

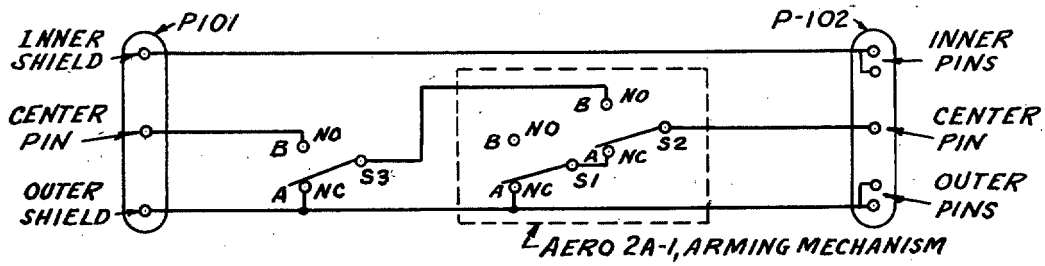
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FIGURE 1

ELECTRICAL CIRCUIT AND OPERATION SEQUENCE

ELECTRICAL FUZE ARMING MECHANISM, AERO 2A-1

BOMB RACK AND ROCKET LAUNCHER, COMBINATION, AERO 15



S_1 and S_2 - Microswitches in the Arming Mechanism. See BuAer Dr. No. 59A7D1

S_3 - Interlock Switch. See Drawing No. 58A154D16

P-101 - Triaxial Cable Connector. See Drawing No. 59A5A125-1

P-102 - Molded Plug Connector. See Drawing No. 59A7D4

OPERATIONAL CONFIGURATION	SWITCH	POSITION
CORD RETRACTED AND HOOKS CLOSED	S_1	A Normally Closed (NC)
	S_2	A Normally Closed (NC)
	S_3	A Normally Closed (NC)
CORD RETRACTED AND HOOKS OPEN	S_1	A (NC)
	S_2	A (NC)
	S_3	B (NO) Normally Open
CORD EXTENDED 4 INCHES AND HOOKS CLOSED	S_1	B
	S_2	B
	S_3	A

OPERATIONAL CONFIGURATION	SWITCH	POSITION
CORD EXTENDED 4 INCHES AND HOOKS OPEN	S_1	B
	S_2	B
	S_3	B
CORD EXTENDED FULLY AND HOOKS CLOSED	S_1	B
	S_2	A
	S_3	A
CORD EXTENDED FULLY AND HOOKS OPEN	S_1	B
	S_2	A
	S_3	B

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